

BFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Complete

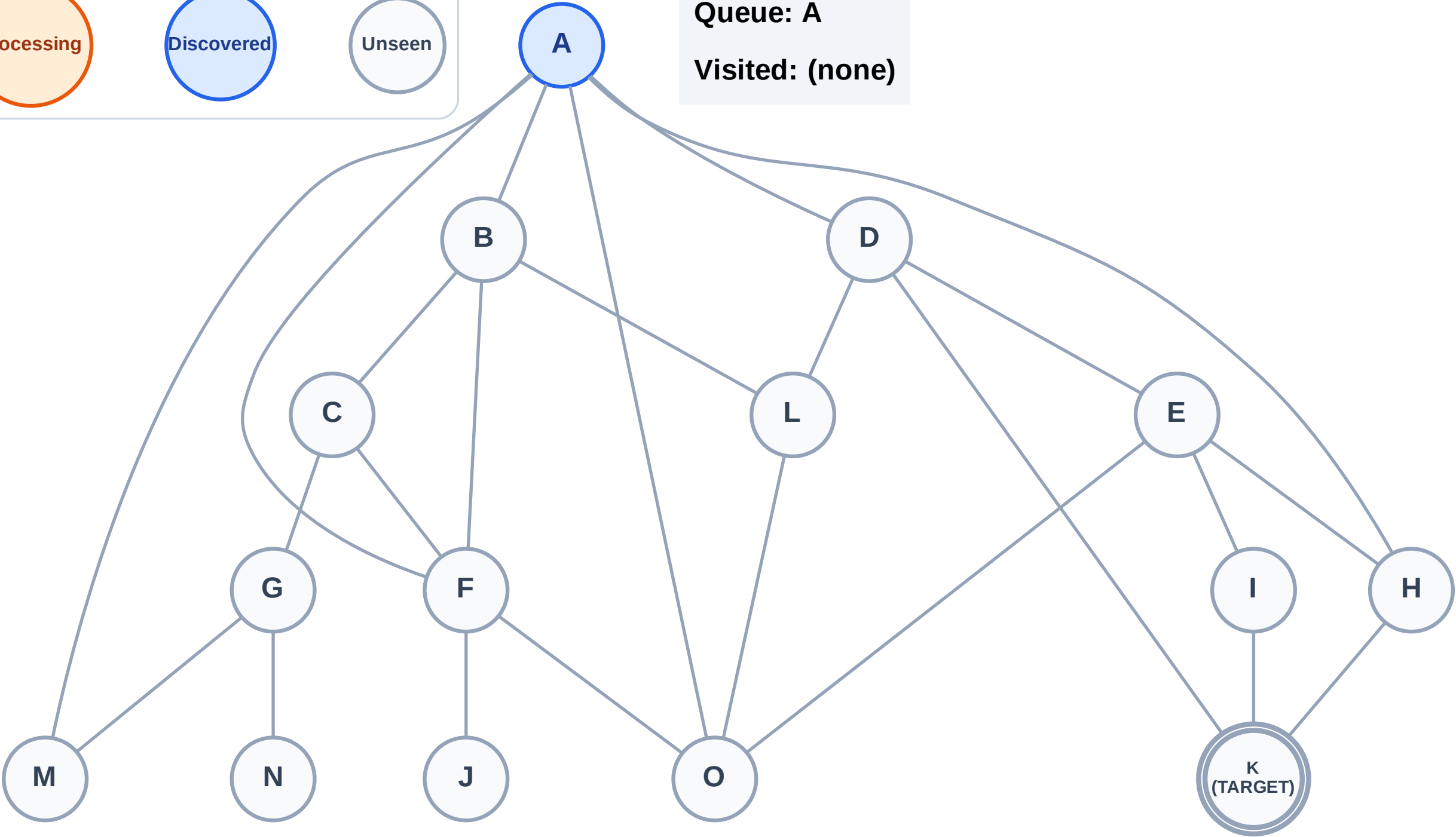
Processing

Discovered

Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

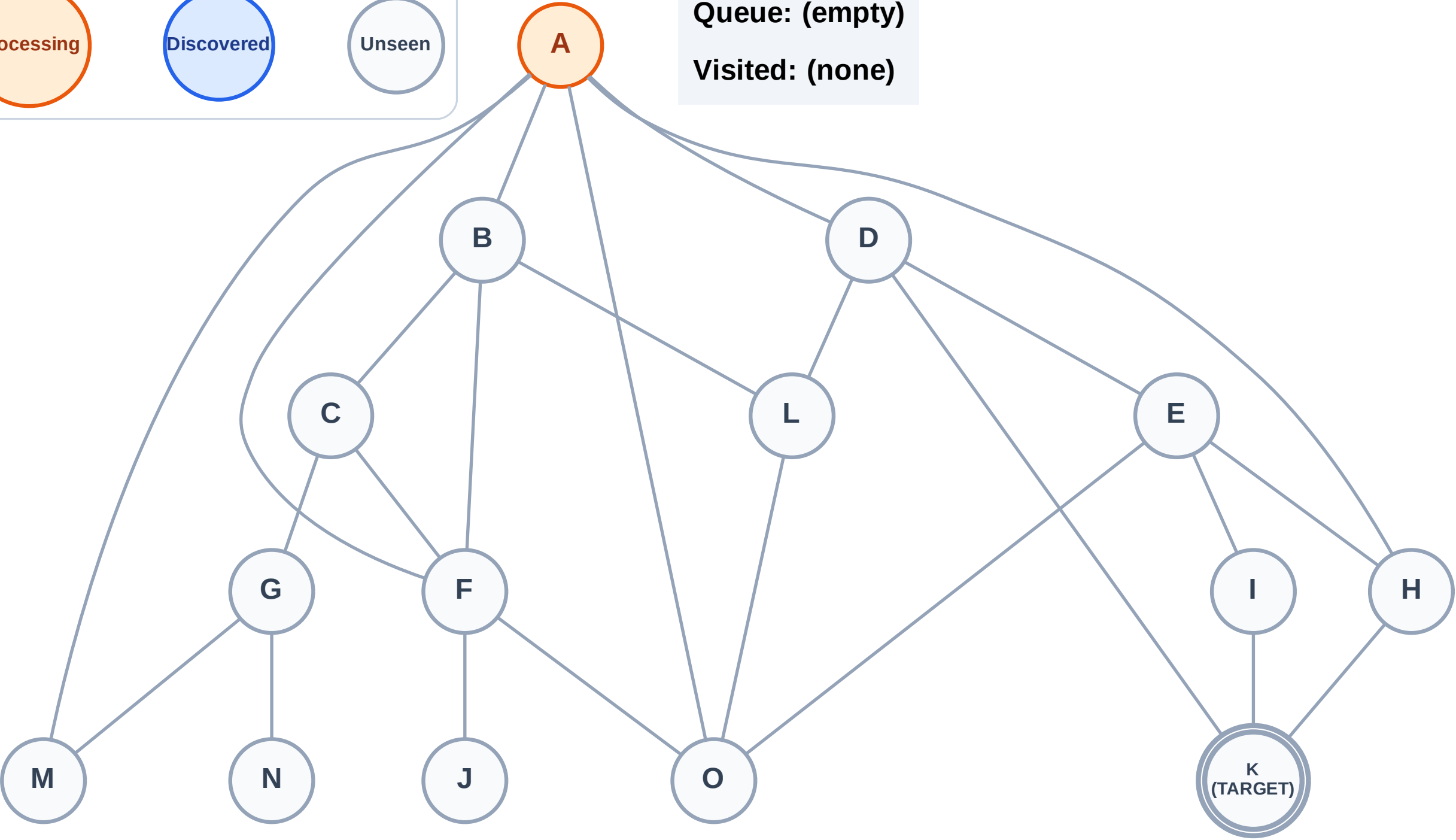
Complete

Processing

Discovered

Unseen

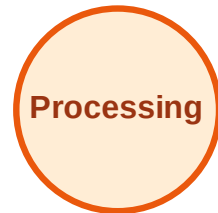
Queue: (empty)
Visited: (none)



BFS Traversal

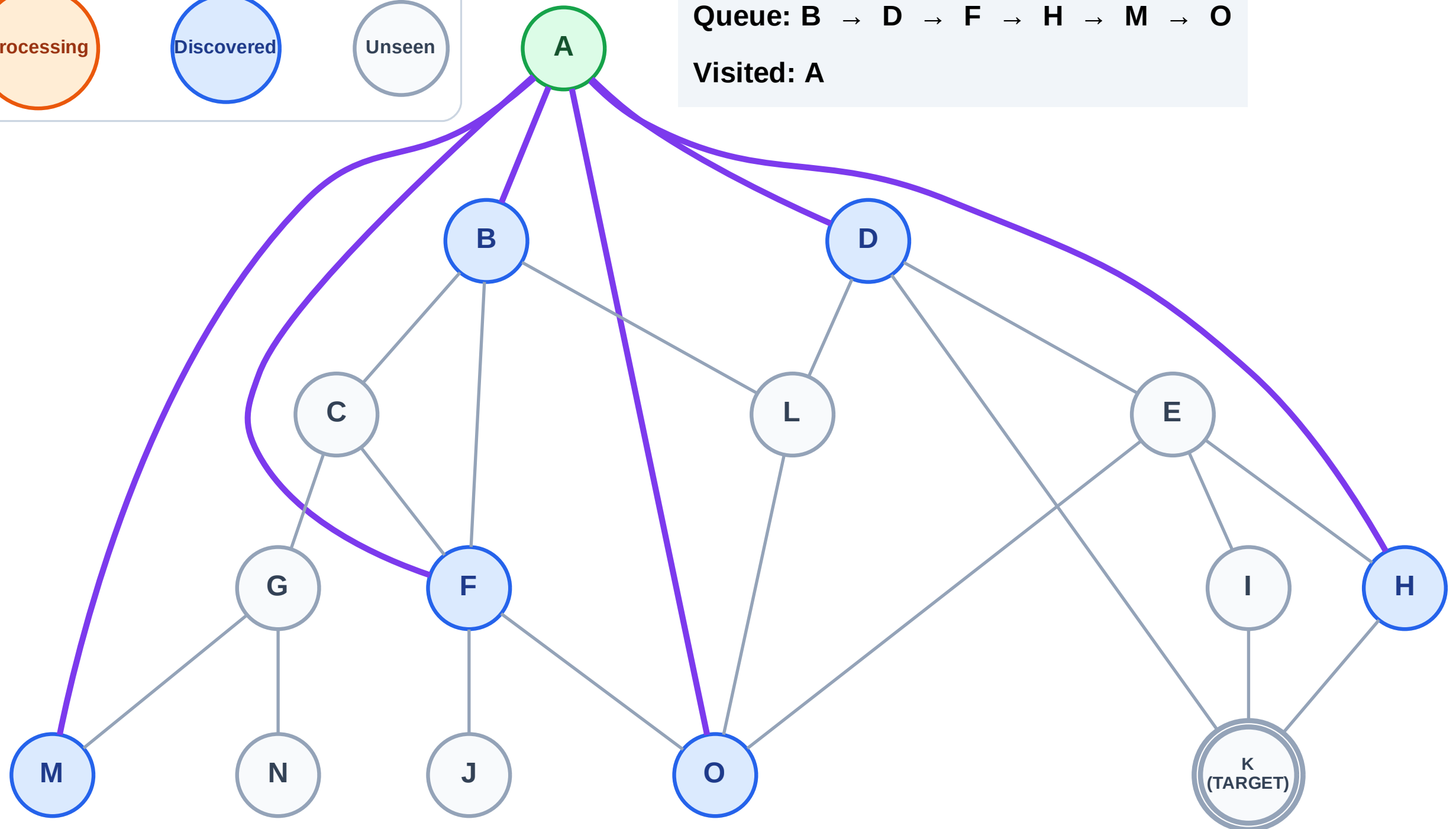
Step 3: Discover B, D, F, H, M, O from A

Legend



Queue: B → D → F → H → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

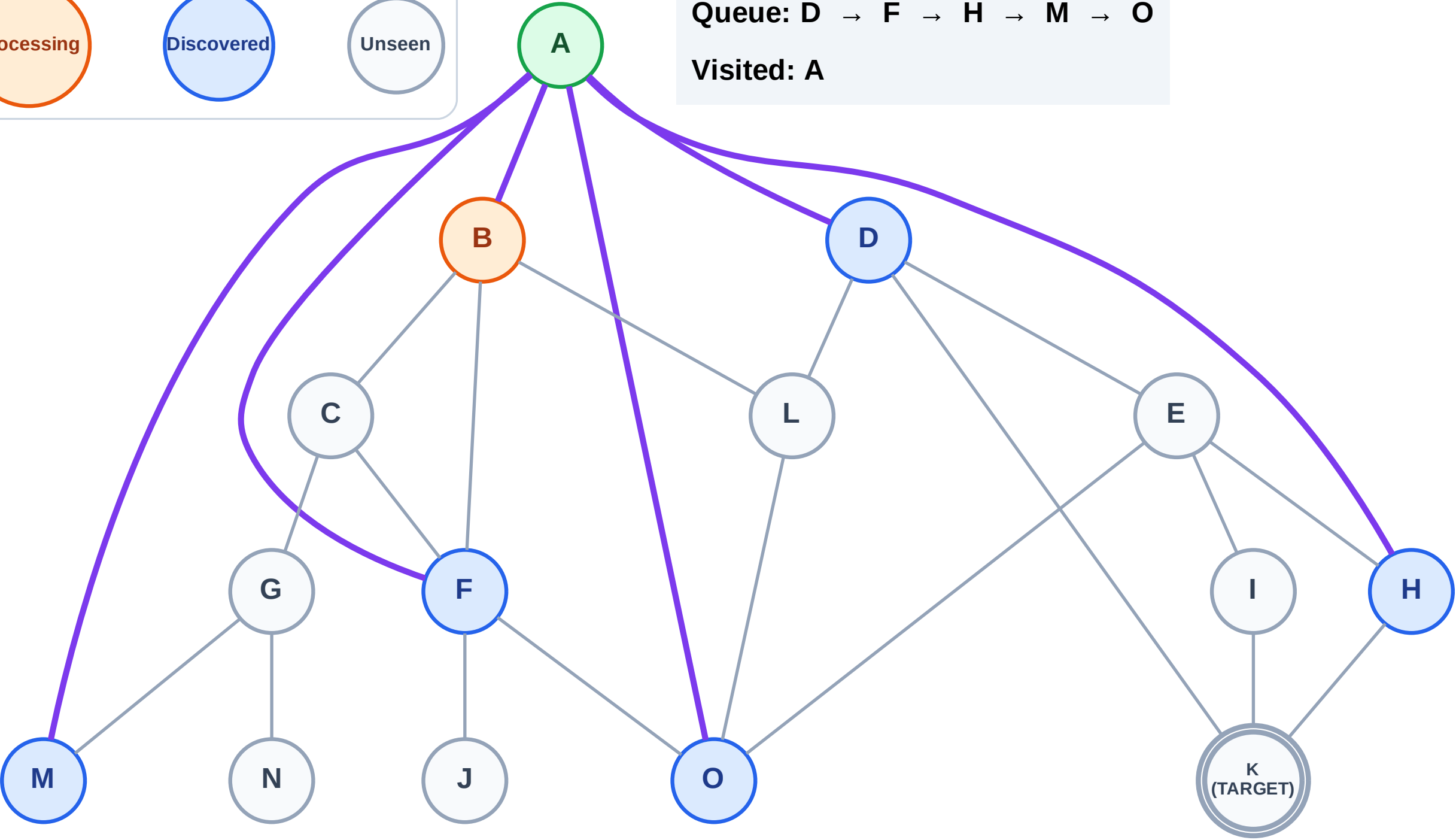
Processing

Discovered

Unseen

Queue: D → F → H → M → O

Visited: A



BFS Traversal

Step 5: Discover C, L from B

Legend

Complete

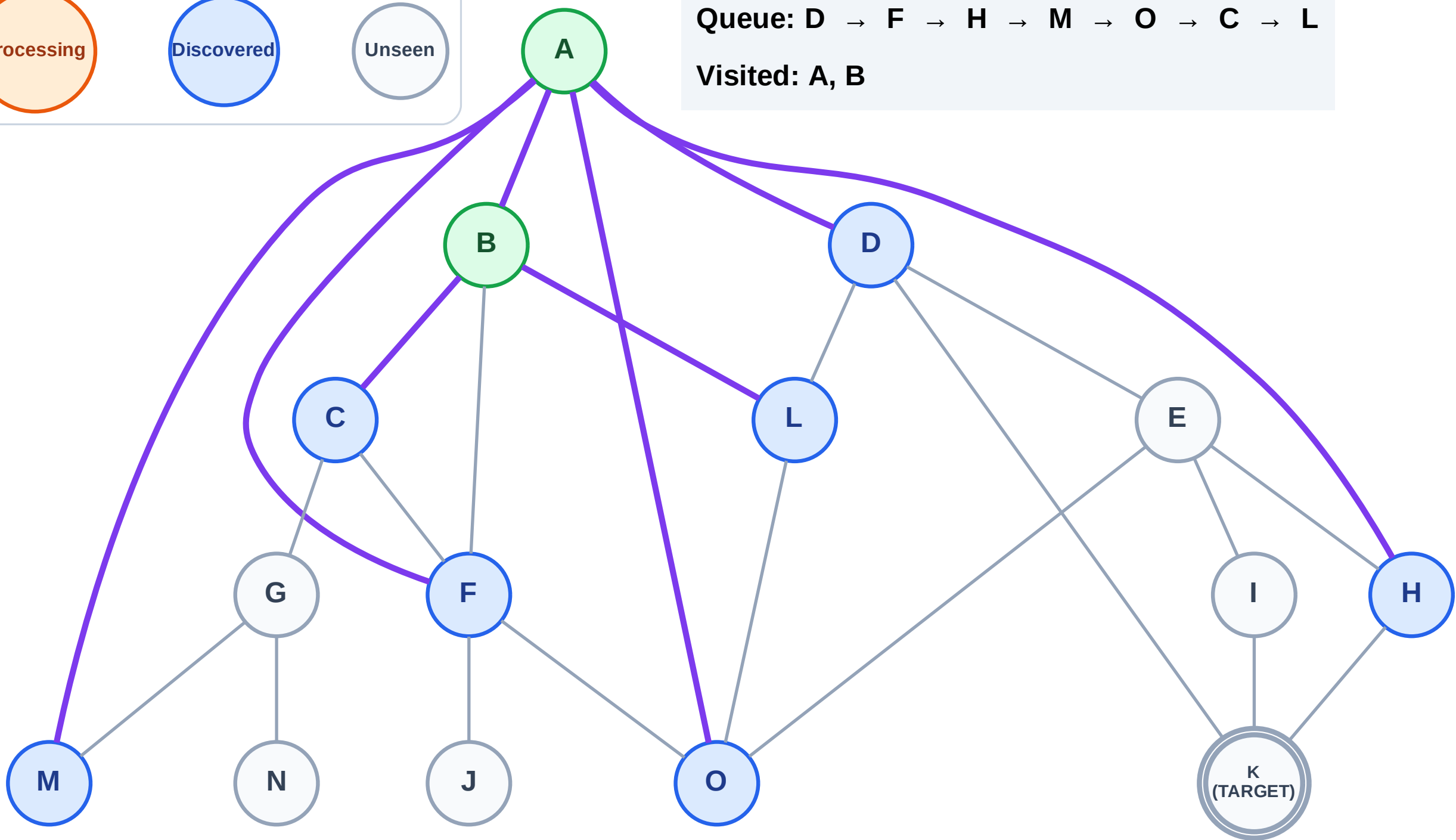
Processing

Discovered

Unseen

Queue: D → F → H → M → O → C → L

Visited: A, B



BFS Traversal

Step 6: Process node D

Legend

Complete

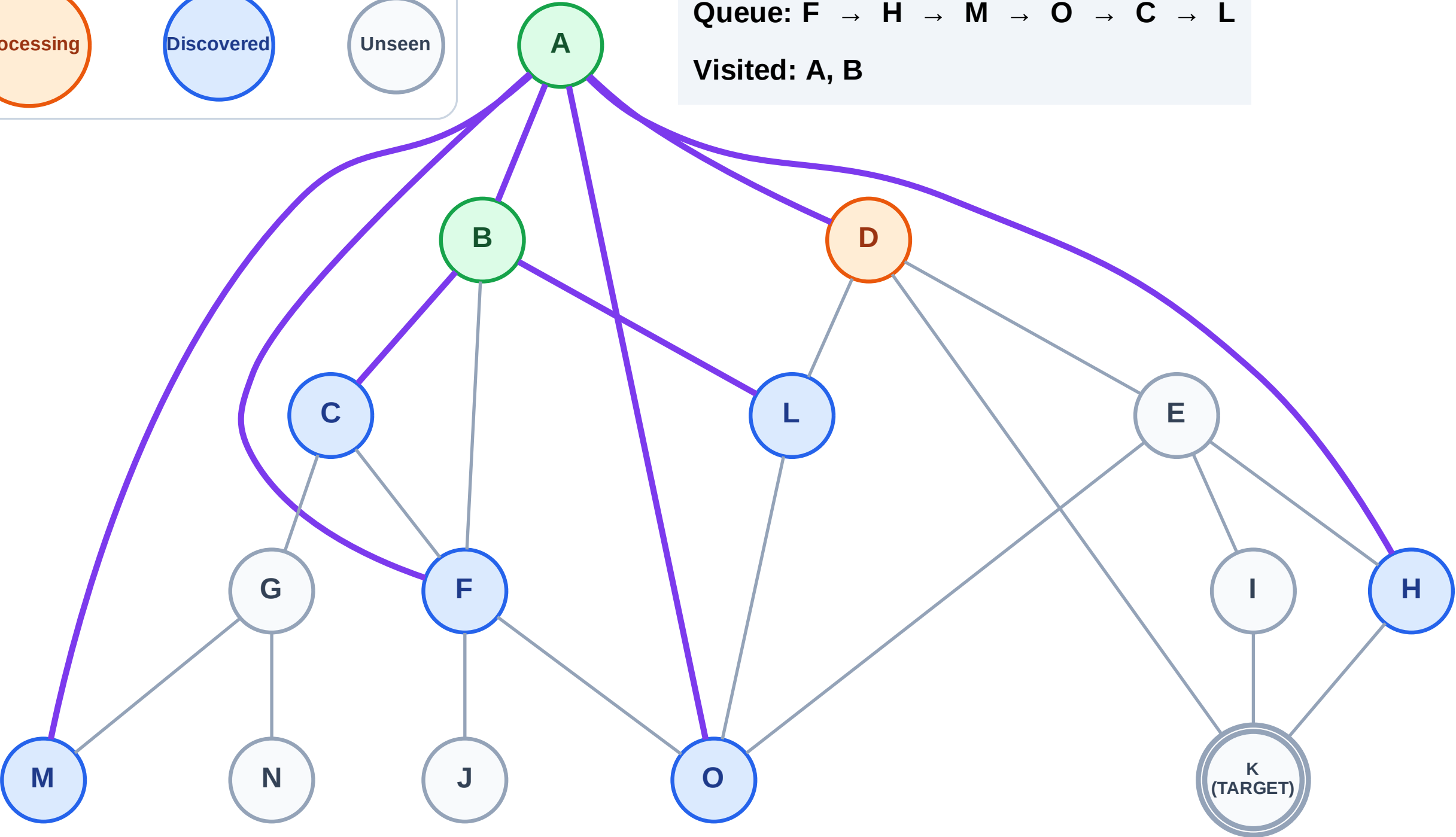
Processing

Discovered

Unseen

Queue: F → H → M → O → C → L

Visited: A, B



BFS Traversal

Step 7: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

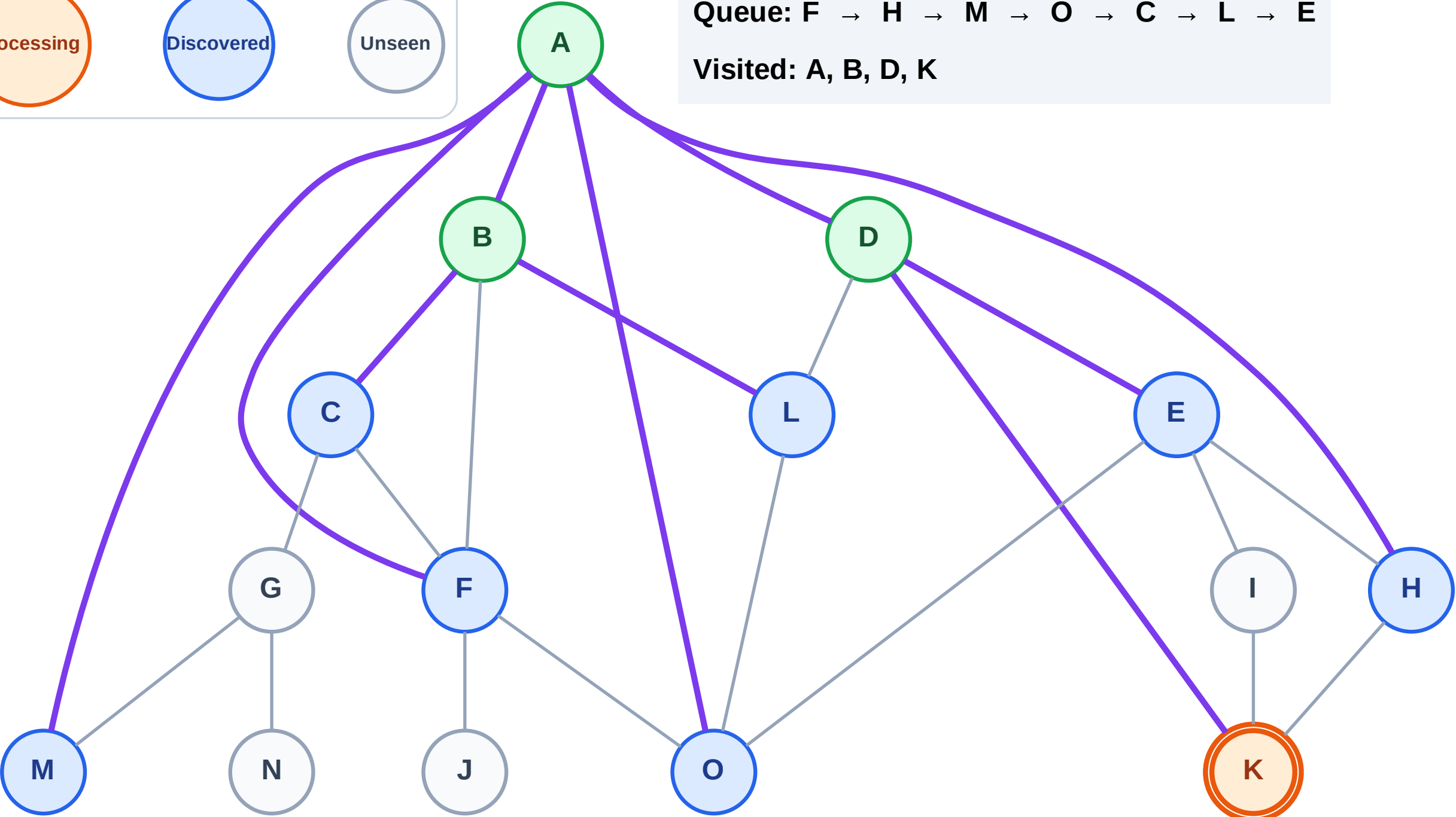
Processing

Discovered

Unseen

Queue: F → H → M → O → C → L → E

Visited: A, B, D, K



DFS Traversal

Step 1: Start at A (Seeking target node K (Eager Short-Circuit))

Legend

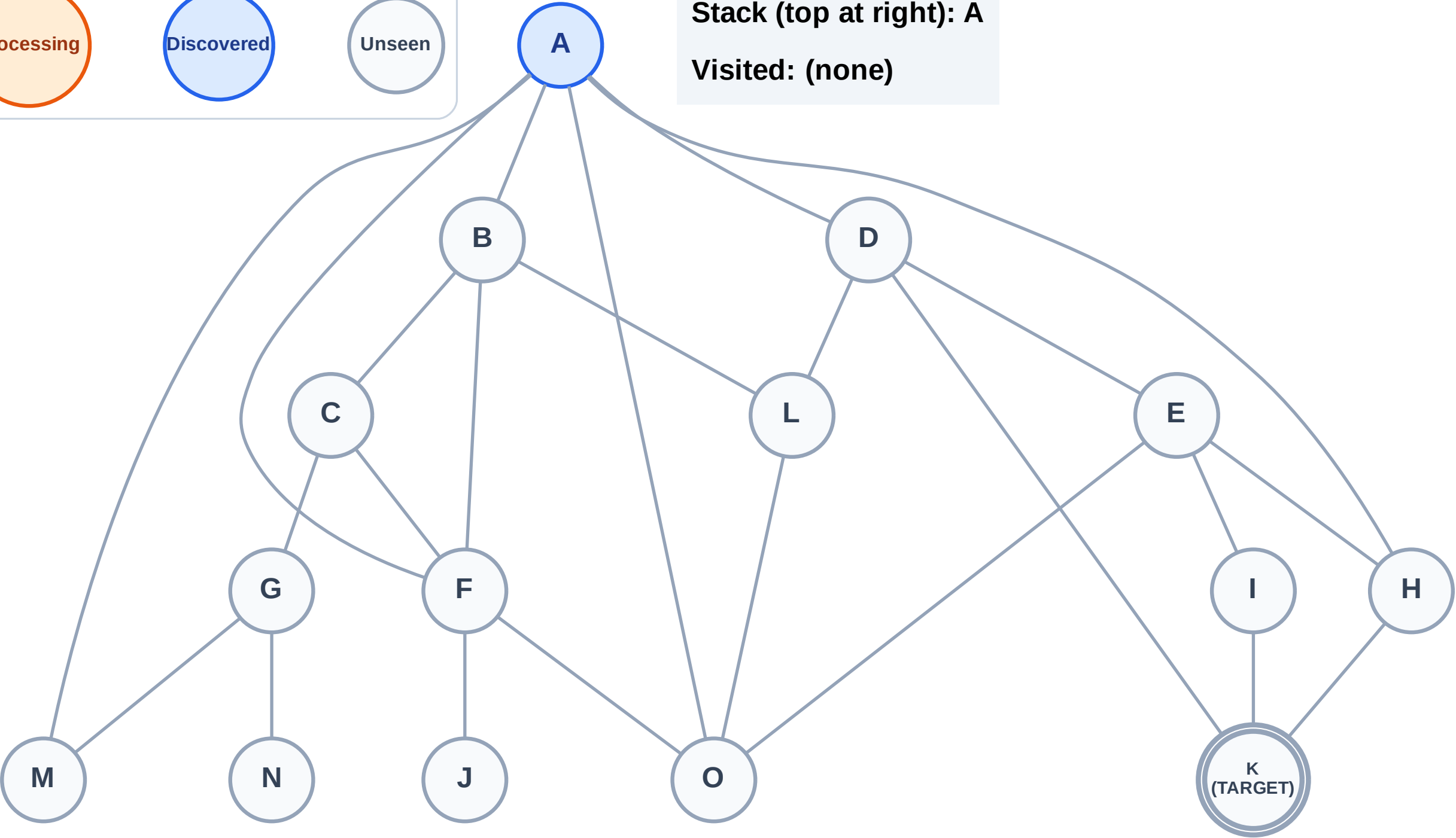
Complete

Processing

Discovered

Unseen

Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

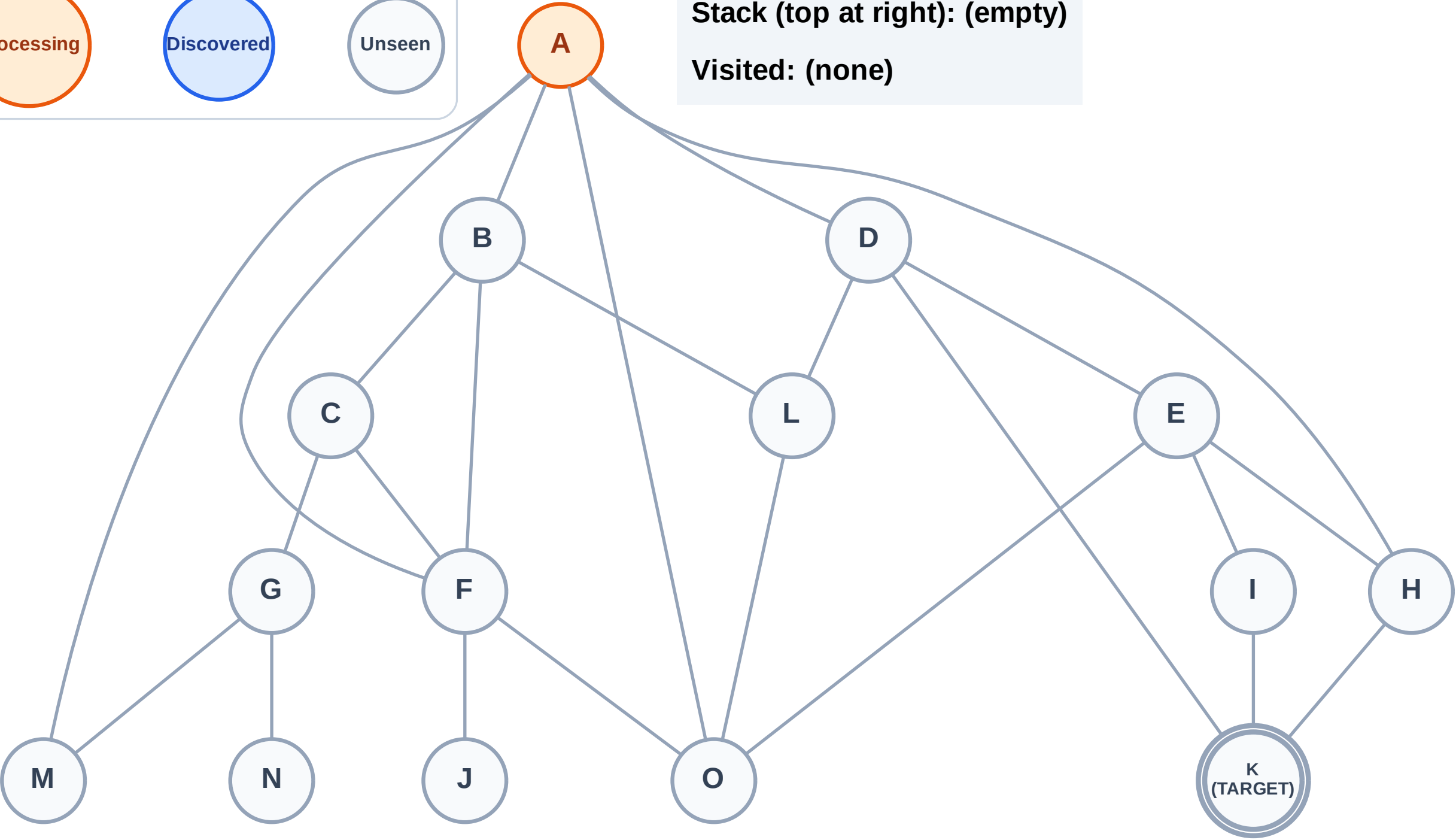
Complete

Processing

Discovered

Unseen

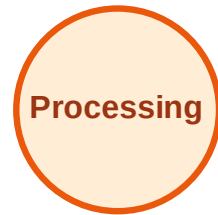
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

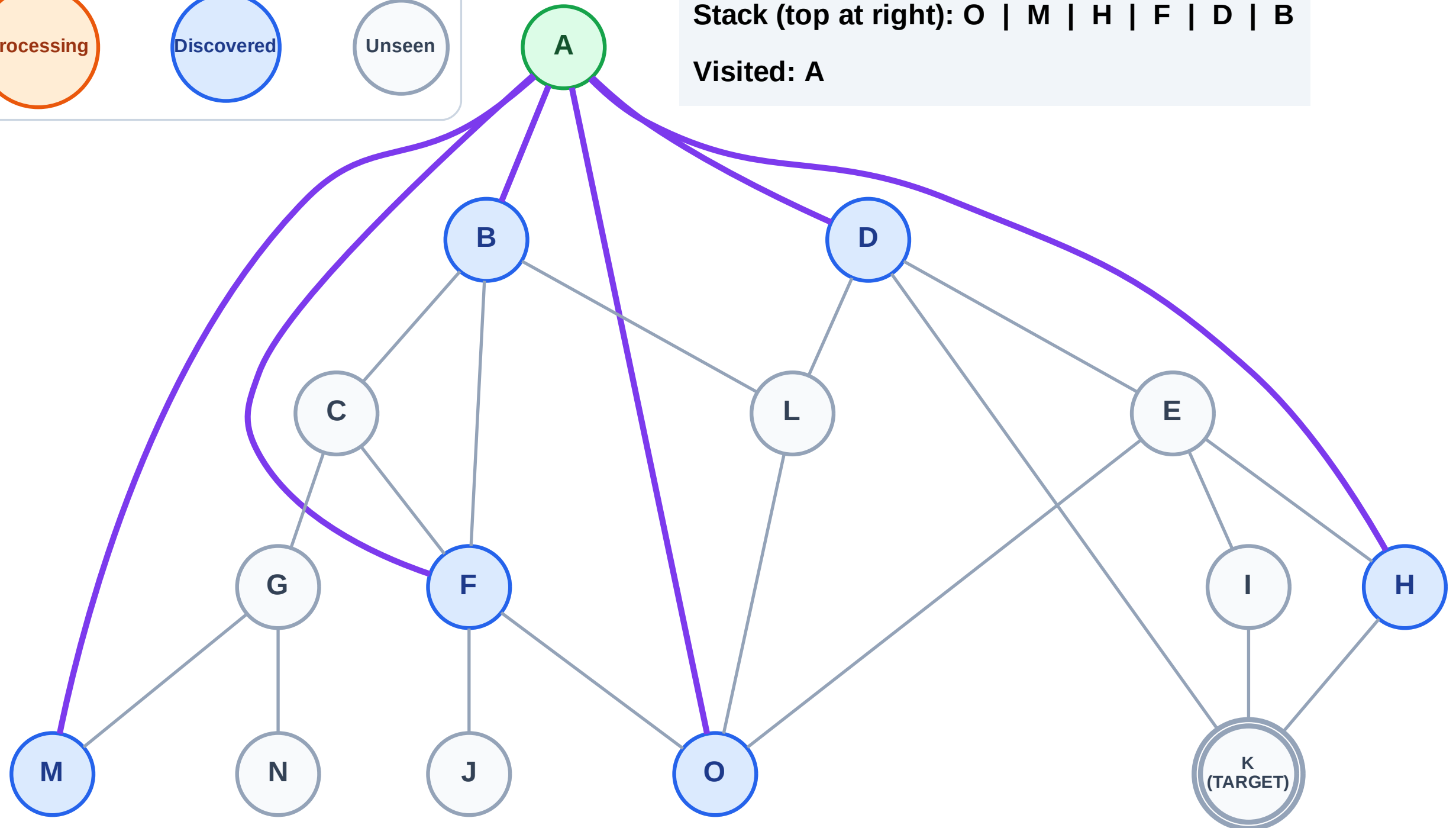
Step 3: Discover O, M, H, F, D, B from A

Legend



Stack (top at right): O | M | H | F | D | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

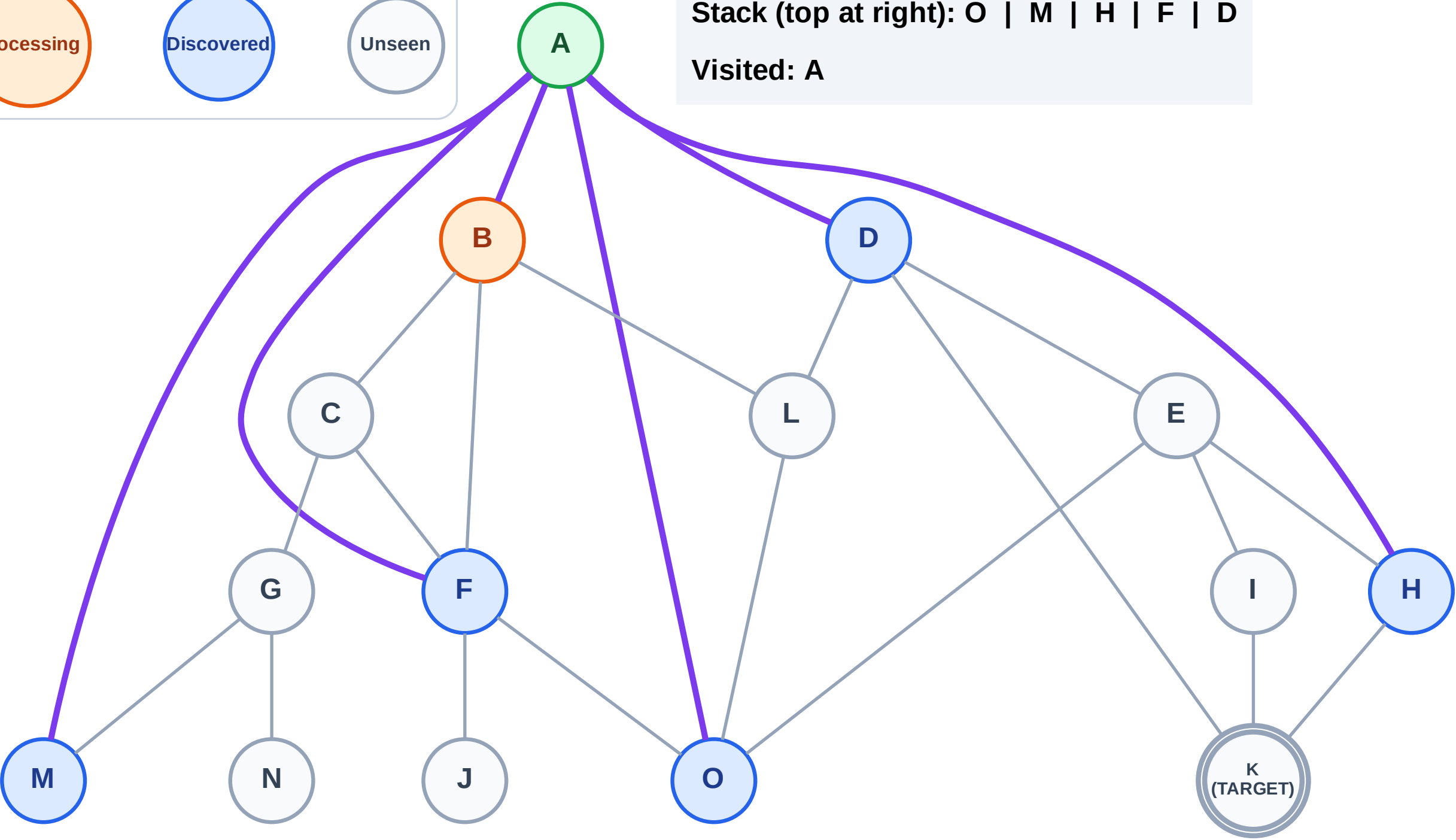
Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D

Visited: A



DFS Traversal

Step 5: Discover L, C from B

Legend

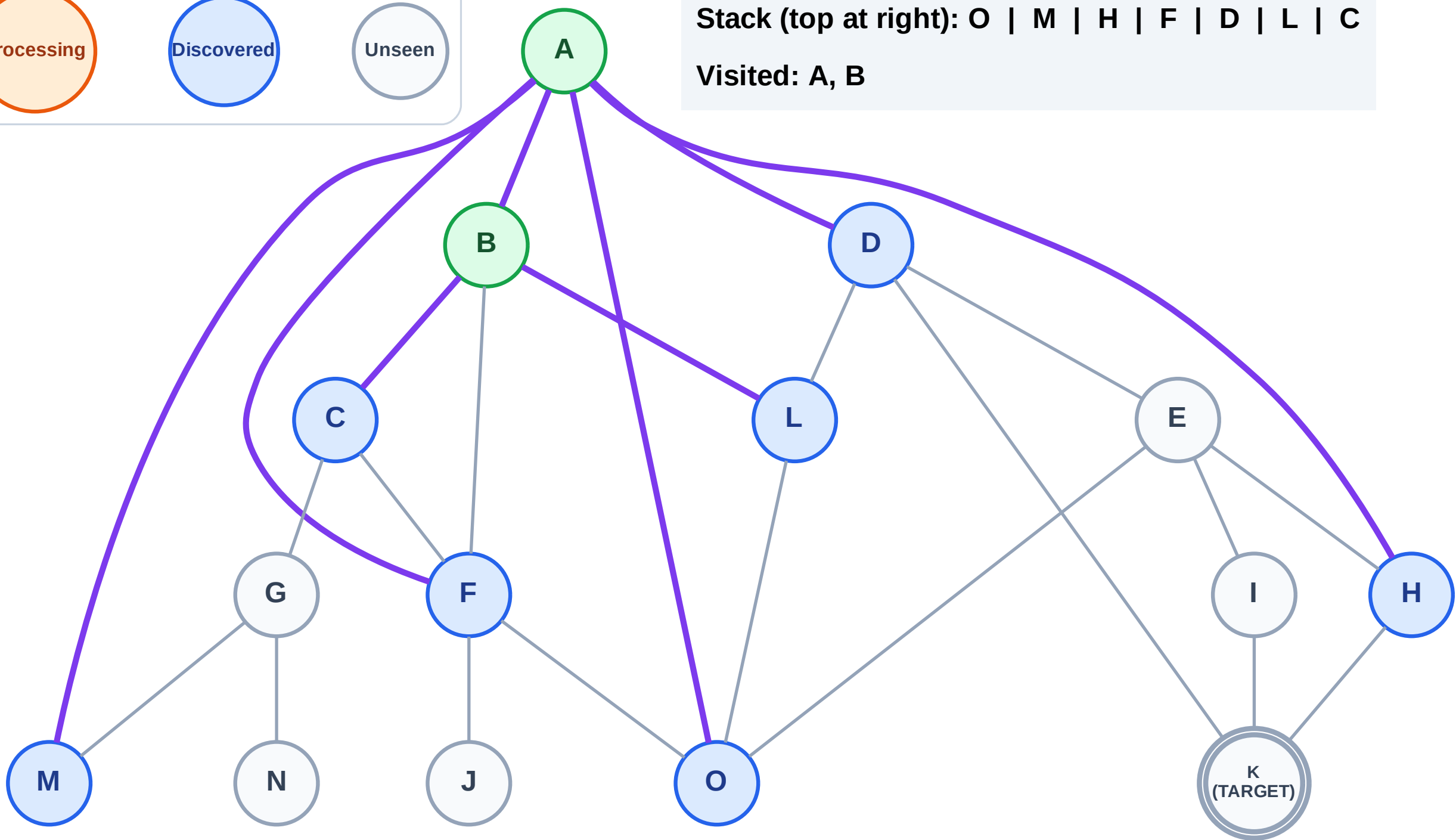
Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D | L | C
Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

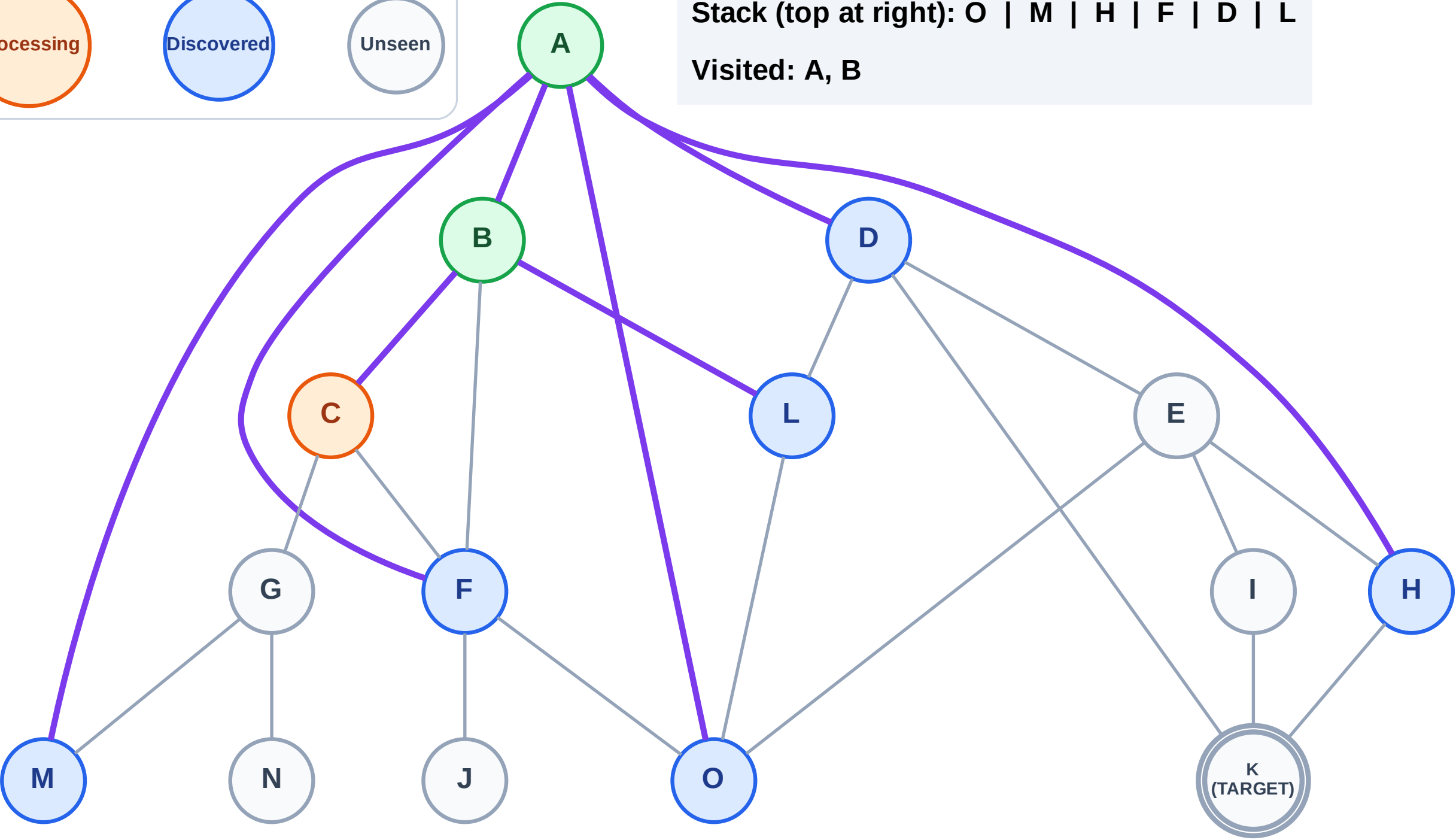
Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D | L

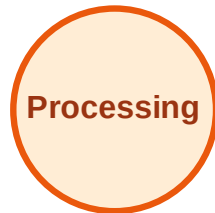
Visited: A, B



DFS Traversal

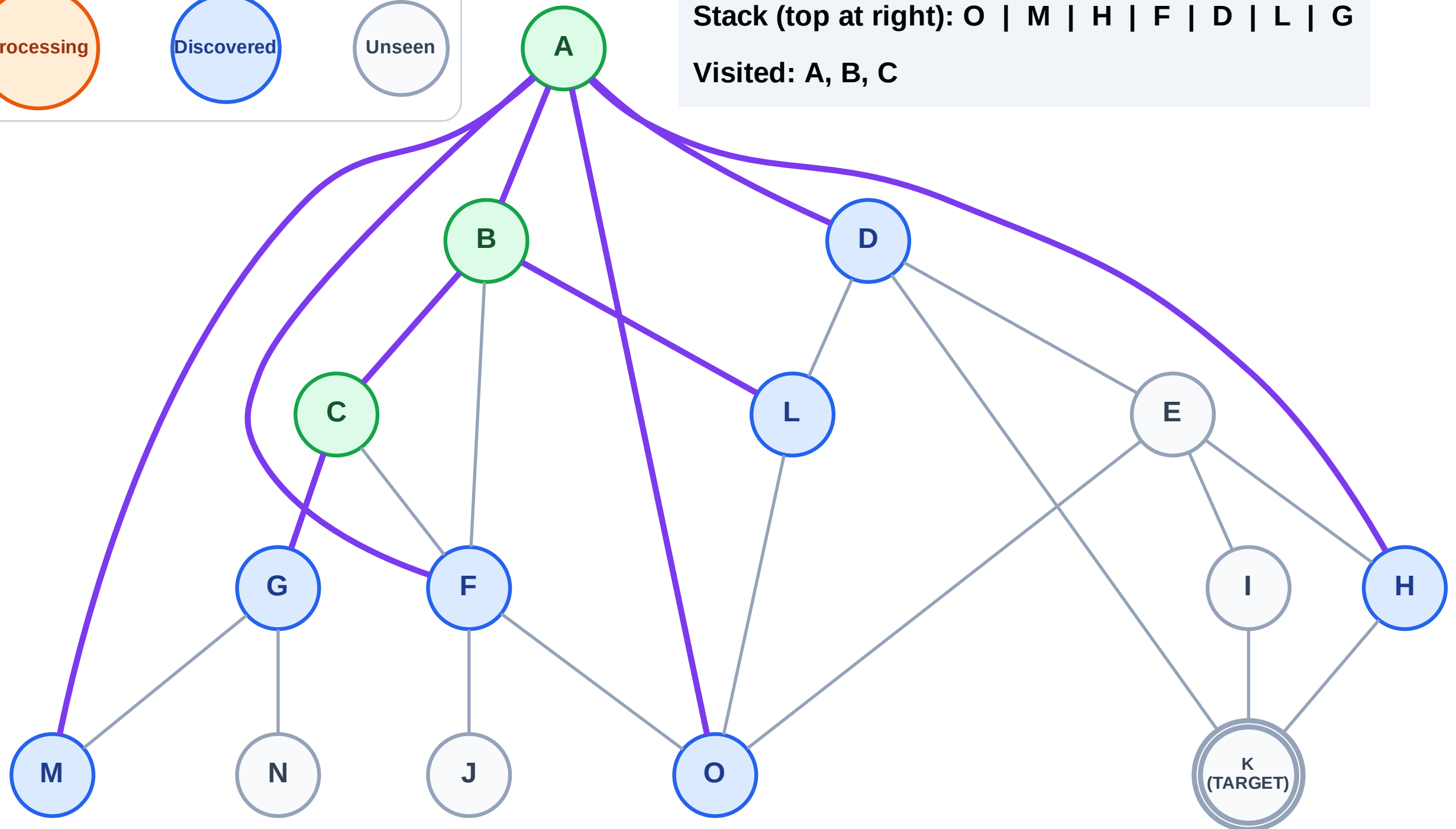
Step 7: Discover G from C

Legend



Stack (top at right): O | M | H | F | D | L | G

Visited: A, B, C



DFS Traversal

Step 8: Process node G

Legend

Complete

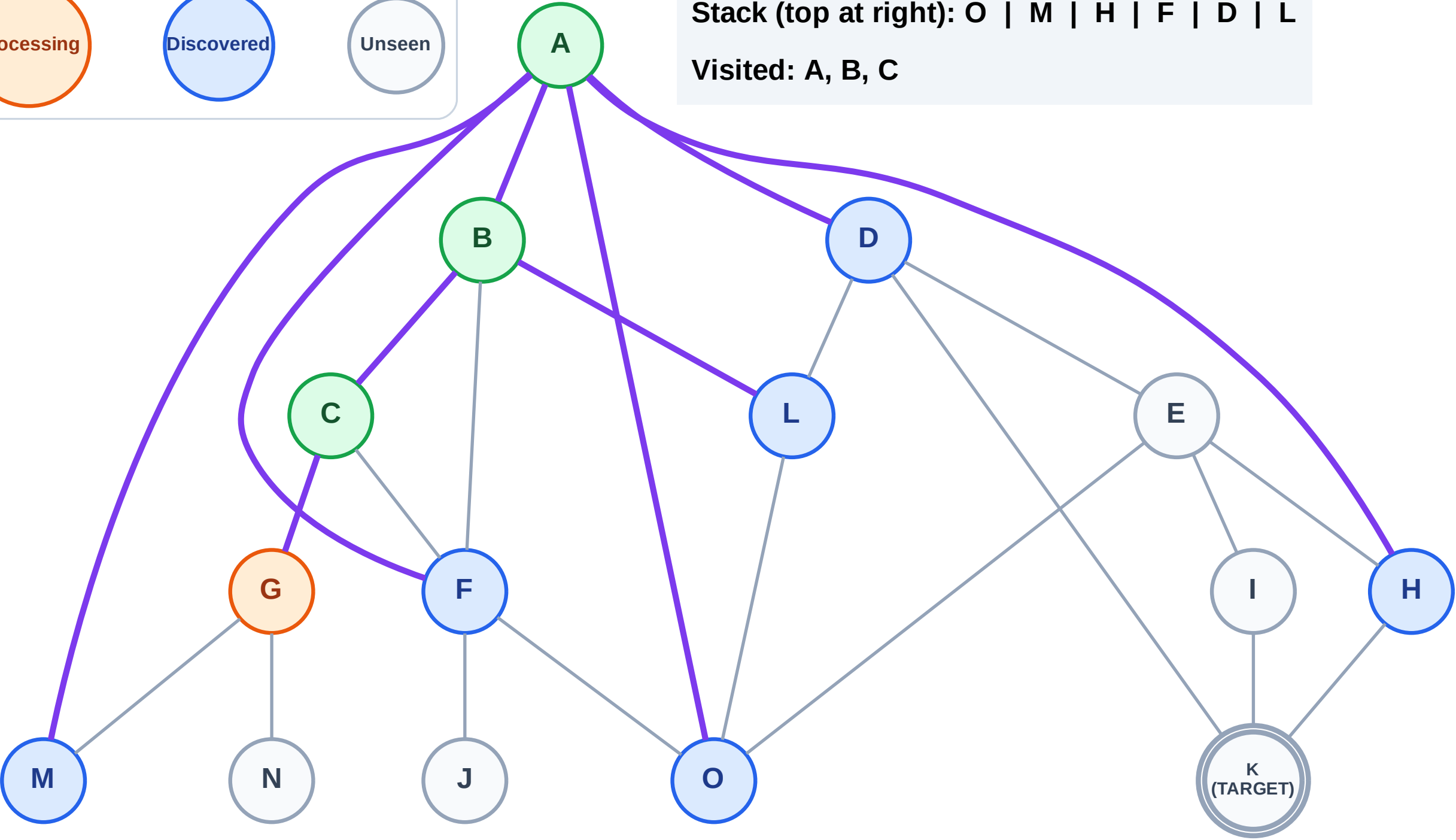
Processing

Discovered

Unseen

Stack (top at right): O | M | H | F | D | L

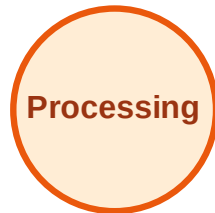
Visited: A, B, C



DFS Traversal

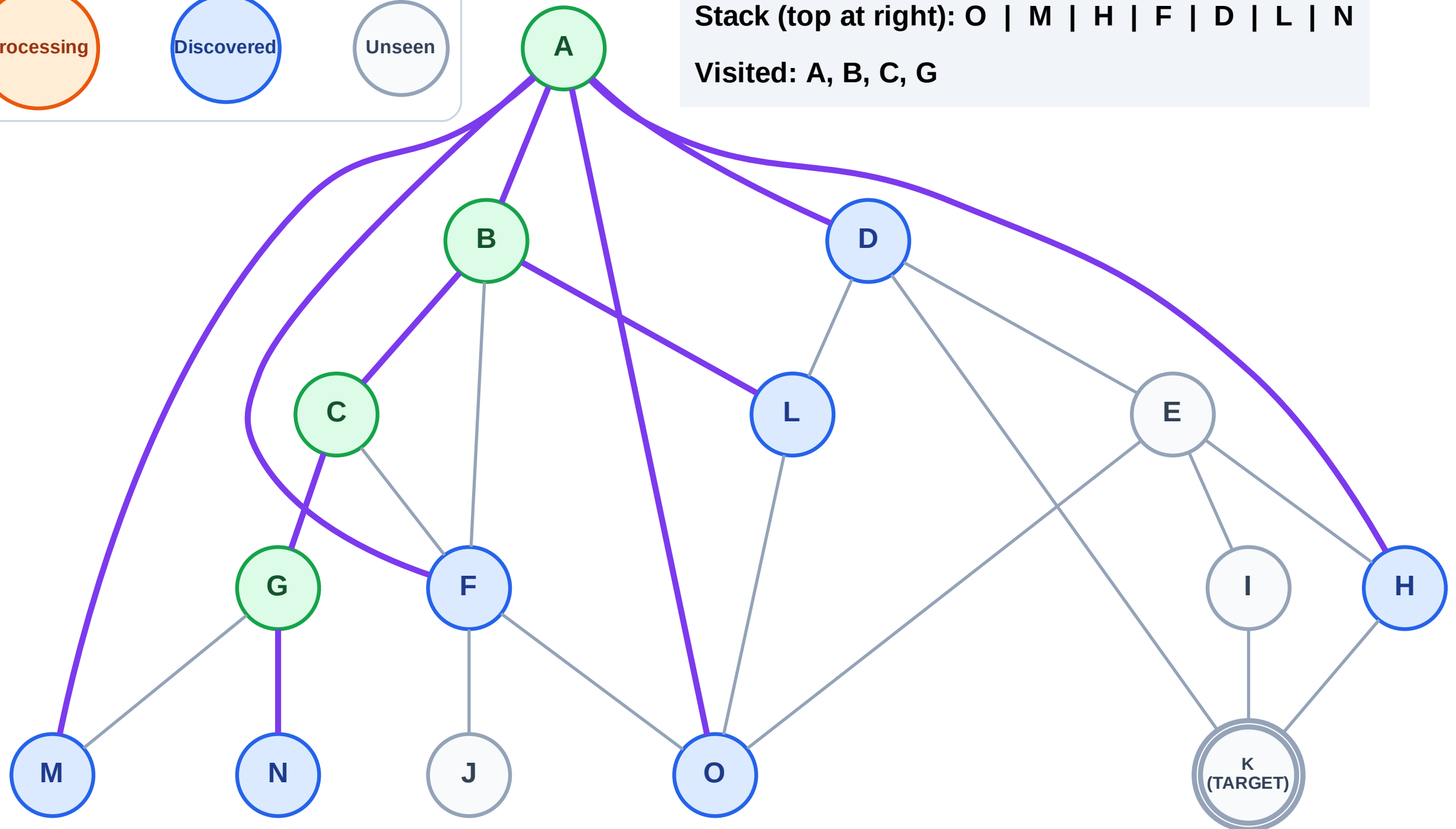
Step 9: Discover N from G

Legend



Stack (top at right): O | M | H | F | D | L | N

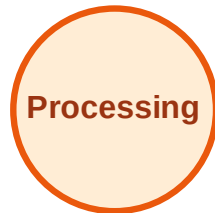
Visited: A, B, C, G



DFS Traversal

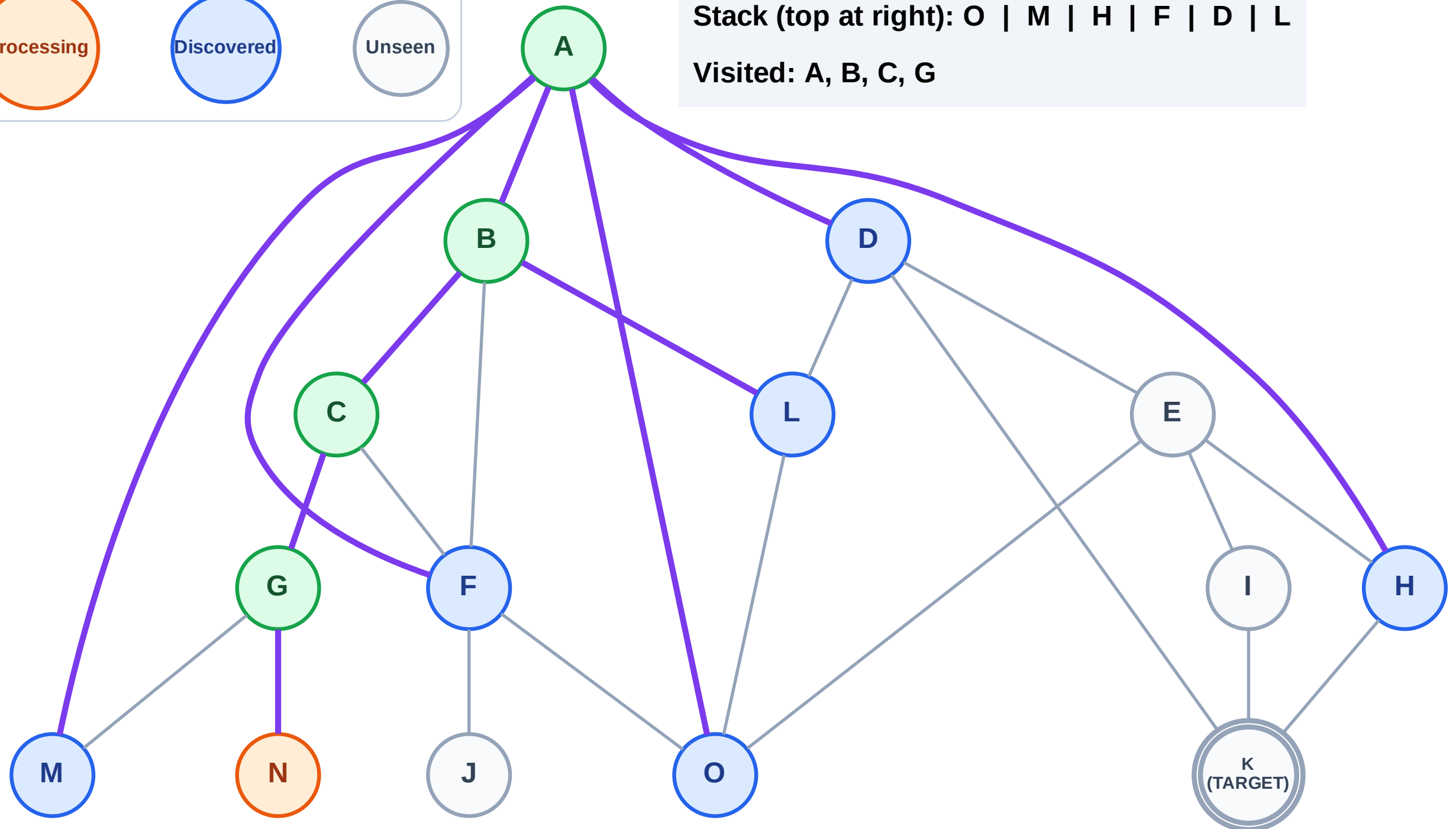
Step 10: Process node N

Legend



Stack (top at right): O | M | H | F | D | L

Visited: A, B, C, G



DFS Traversal

Step 11: No new neighbors from N

Legend

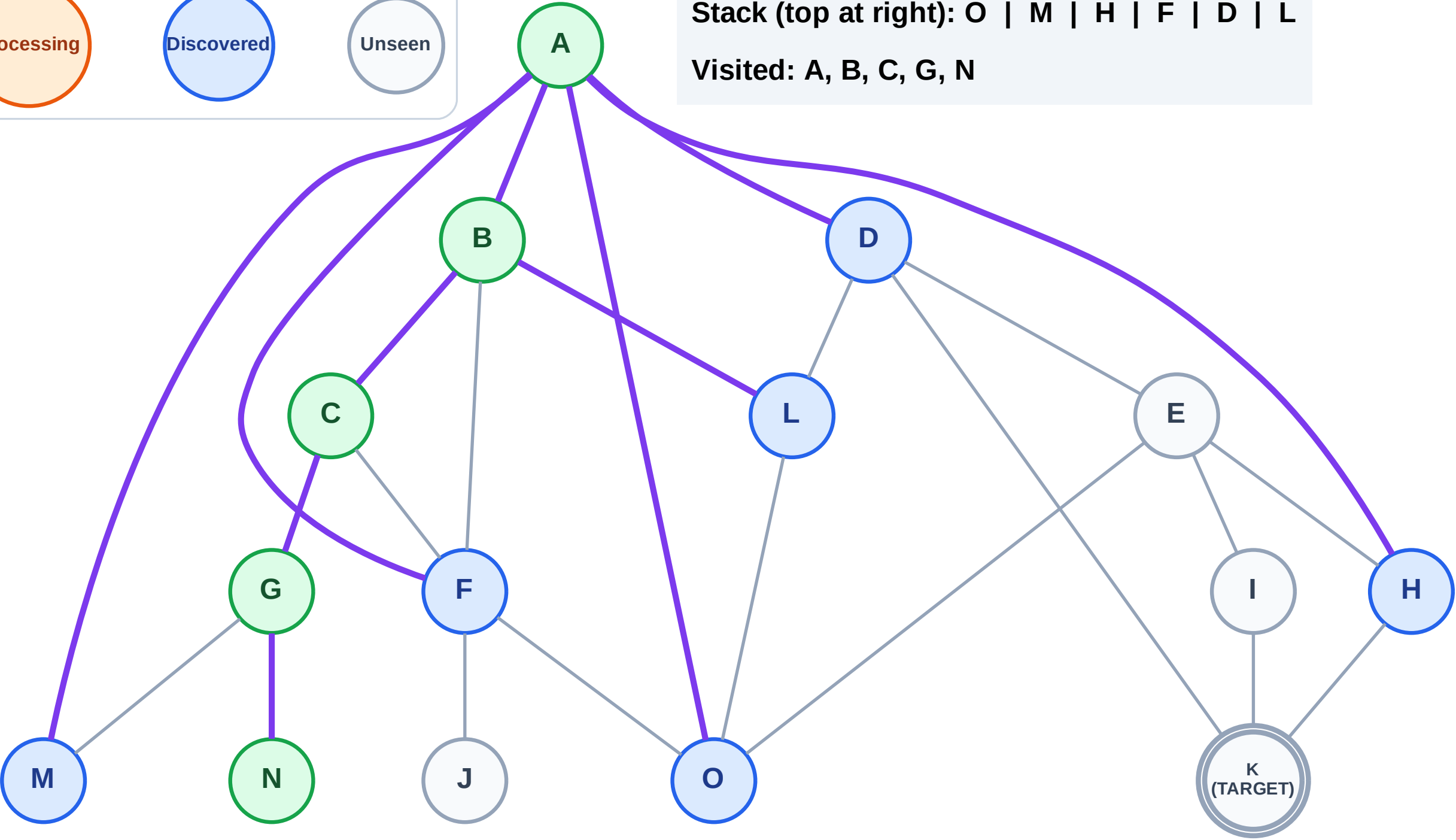
Complete

Processing

Discovered

Unseen

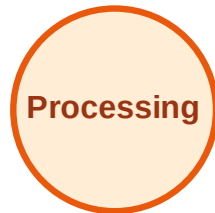
Stack (top at right): O | M | H | F | D | L
Visited: A, B, C, G, N



DFS Traversal

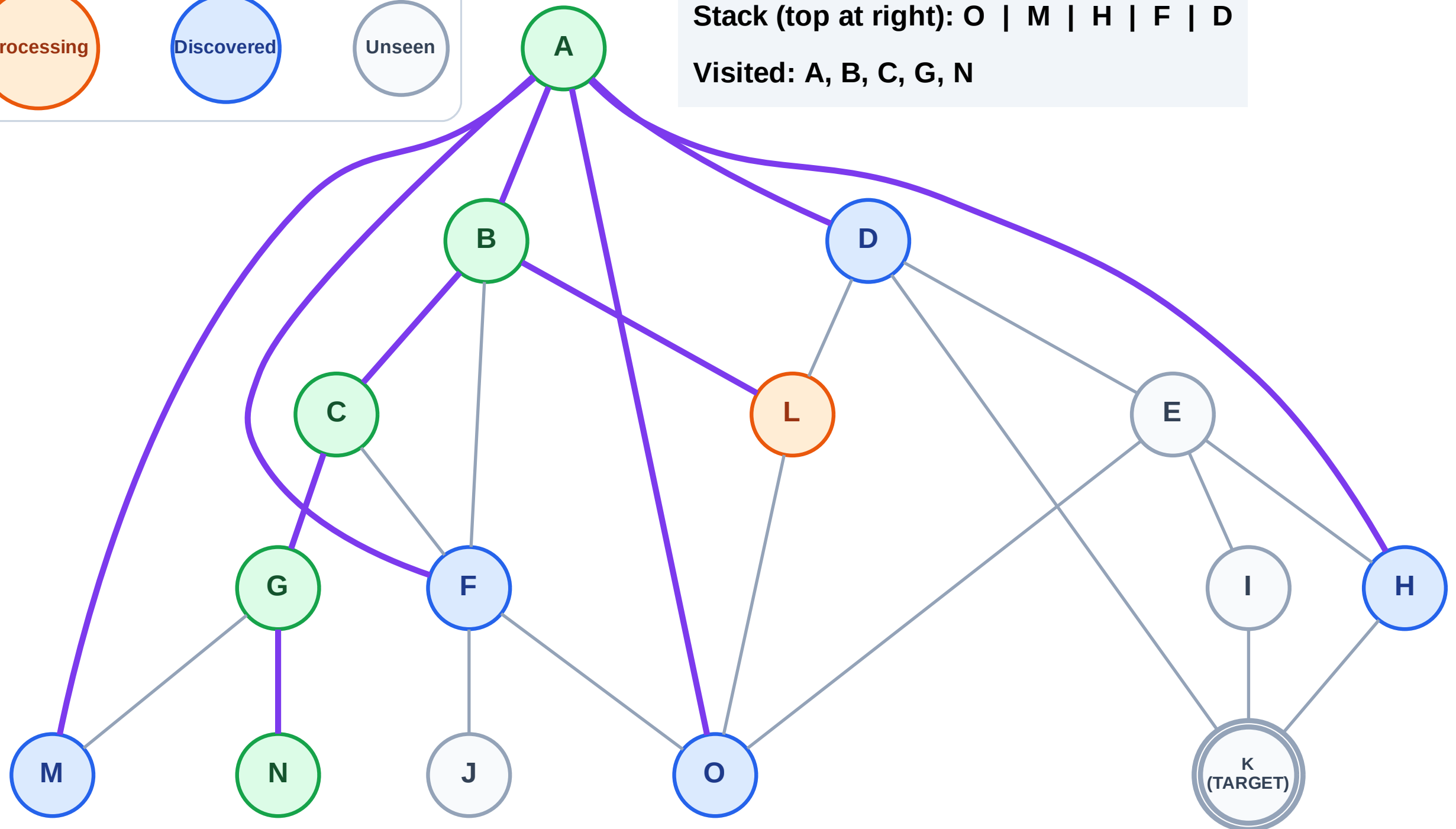
Step 12: Process node L

Legend



Stack (top at right): O | M | H | F | D

Visited: A, B, C, G, N



DFS Traversal

Step 13: No new neighbors from L

Legend

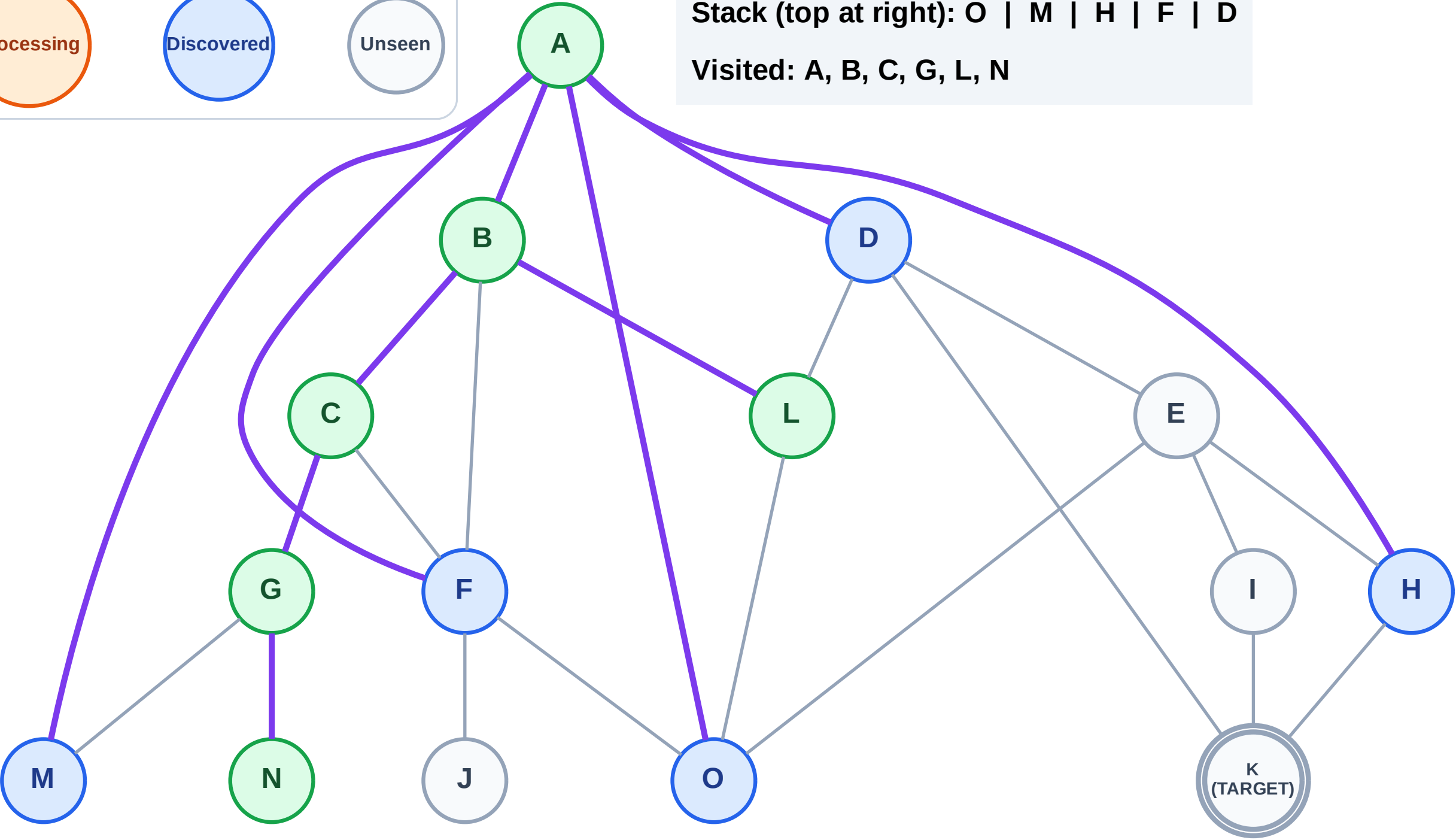
Complete

Processing

Discovered

Unseen

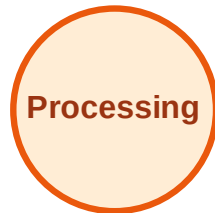
Stack (top at right): O | M | H | F | D
Visited: A, B, C, G, L, N



DFS Traversal

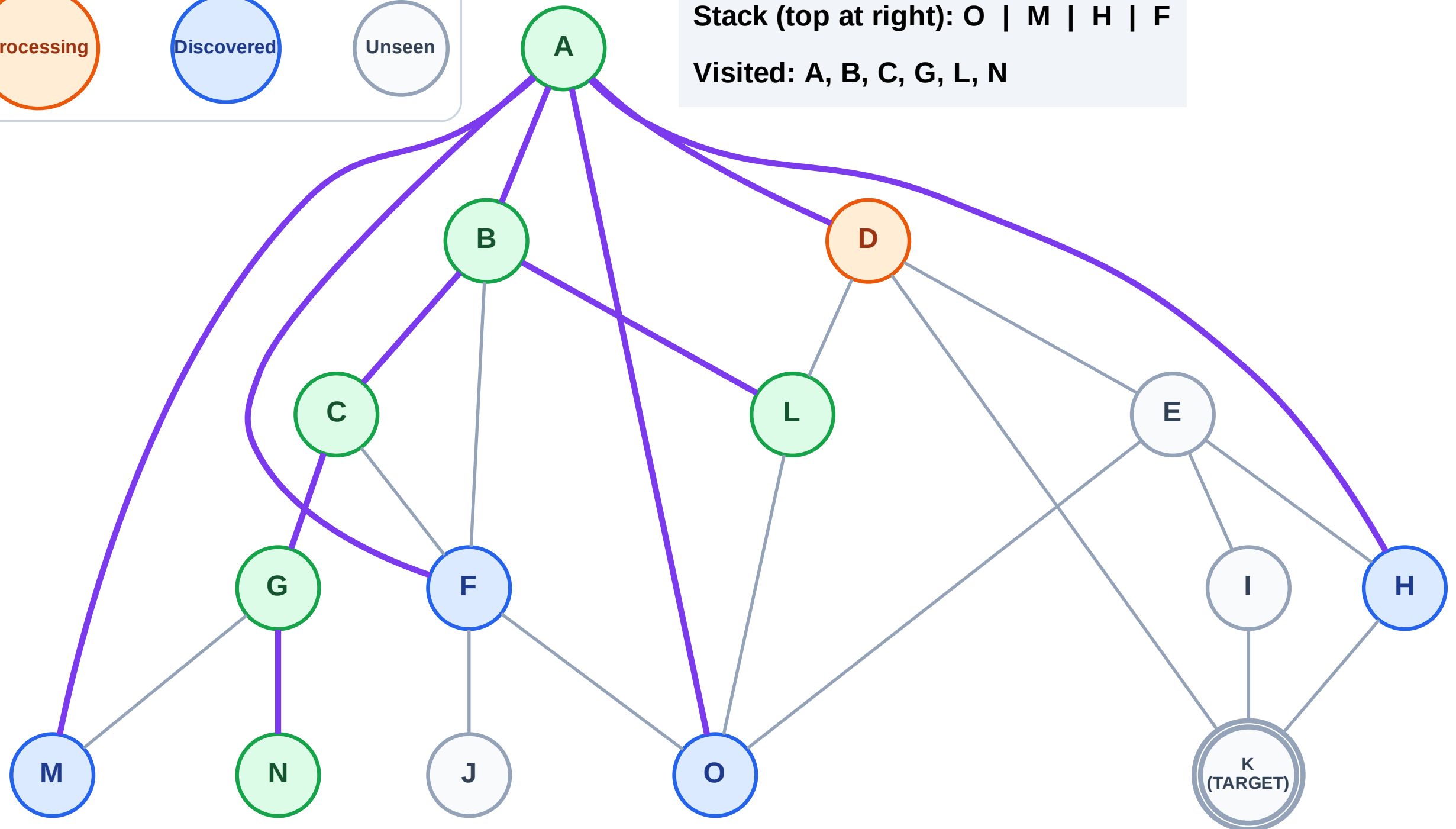
Step 14: Process node D

Legend



Stack (top at right): O | M | H | F

Visited: A, B, C, G, L, N



DFS Traversal

Step 15: TARGET NODE K FOUND (SHORT-CIRCUITED)! Path (2 hops): A → D → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): O | M | H | F
Visited: A, B, C, D, G, K, L, N

